

Completion of arbitrary b -metric spaces

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Source and target

The source paper is S. Cobzas, *b-metric spaces, fixed points and Lipschitz functions*, arXiv:1802.02722 [1]. A b -metric on X is a symmetric map $d : X \times X \rightarrow [0, \infty)$ with $d(x, y) = 0$ iff $x = y$, satisfying the relaxed triangle inequality

$$d(x, z) \leq s(d(x, y) + d(y, z))$$

for some $s \geq 1$. A completion means a complete b -metric space (Y, ρ) containing an isometric dense copy of X .

Remark 1.18 of the source paper states the open problem:

as a **QUESTION**.

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Result

Theorem 1. *Every b -metric space admits a completion. More precisely, if (X, d) has relaxed triangle constant s , then there exist a complete b -metric space $(\hat{X}, D)$, also with relaxed triangle constant s , and an isometric embedding $j : X \rightarrow \hat{X}$ such that $j(X)$ is dense in $\hat{X}$.*

Remark 1. *The construction below proves existence. It deliberately does not claim a canonical completion or uniqueness up to isometry. The missing continuity of an arbitrary b -metric is precisely what prevents the usual canonical “limit of distances” construction.*

Proof Intuition

For strong b -metrics, the standard Cauchy-sequence quotient works because the number $\lim_n d(x_n, y_n)$ is independent of representatives. For a general b -metric this limit may fail to exist or may change when the representatives are changed.

The key move is to give up canonicity. Form the usual equivalence classes of Cauchy sequences, but choose one representative sequence for each class. For old points, choose the constant representative. Then define the distance between two classes by a synchronized limsup of the chosen representatives:

$$D(\xi, \eta) = \limsup_{n \rightarrow \infty} d(x_n^\xi, x_n^\eta).$$

Because the same index n is used on both sides, the relaxed triangle inequality passes directly through the limsup. Constant representatives preserve the original distances exactly, and a diagonal argument gives completeness.

Proof

Let (X, d) be a b -metric space with relaxed triangle constant s . Let $\mathcal{C}$ be the set of all d -Cauchy sequences in X . For $\mathbf{x} = (x_n)$ and $\mathbf{y} = (y_n)$ in $\mathcal{C}$, put

$$\mathbf{x} \sim \mathbf{y} \iff d(x_n, y_n) \rightarrow 0.$$

This is an equivalence relation. Reflexivity and symmetry are immediate, and transitivity follows from the relaxed triangle inequality.

Let $\widehat{X} = \mathcal{C}/\sim$. For each class $\xi \in \widehat{X}$, choose one representative sequence

$$x^\xi = (x_n^\xi)_{n \geq 1} \in \xi.$$

For the class of a constant sequence $c_x = (x, x, \dots)$, choose c_x itself. Define

$$D(\xi, \eta) = \limsup_{n \rightarrow \infty} d(x_n^\xi, x_n^\eta).$$

The value is finite because d is finite-valued.

We first check that D is a b -metric. Symmetry is clear, and $D(\xi, \xi) = 0$. Conversely, if $D(\xi, \eta) = 0$, then $d(x_n^\xi, x_n^\eta) \rightarrow 0$, so the chosen representatives are equivalent and $\xi = \eta$. Finally,

$$d(x_n^\xi, x_n^\zeta) \leq s(d(x_n^\xi, x_n^\eta) + d(x_n^\eta, x_n^\zeta))$$

for every n . Taking limsups gives

$$D(\xi, \zeta) \leq s(D(\xi, \eta) + D(\eta, \zeta)).$$

Thus D is a b -metric with the same relaxed triangle constant s .

Define $j : X \rightarrow \widehat{X}$ by $j(x) = [c_x]$. Since constant representatives were chosen for constant classes,

$$D(j(x), j(y)) = \limsup_n d(x, y) = d(x, y).$$

Hence j is an isometric embedding.

The image $j(X)$ is dense. Let $\xi \in \widehat{X}$ and let (x_n^ξ) be its chosen representative. For fixed N ,

$$D(\xi, j(x_N^\xi)) = \limsup_{n \rightarrow \infty} d(x_n^\xi, x_N^\xi).$$

Because (x_n^ξ) is Cauchy, the right-hand side tends to 0 as $N \rightarrow \infty$.

It remains to prove completeness. Let (ξ_k) be a D -Cauchy sequence in $\widehat{X}$. Choose a decreasing sequence $\varepsilon_j \downarrow 0$ and increasing integers k_j such that

$$D(\xi_m, \xi_n) < \varepsilon_j \quad \text{whenever } m, n \geq k_j.$$

Put $\eta_j = \xi_{k_j}$, and let

$$a^j = (a_n^j)_{n \geq 1}$$

be the chosen representative of η_j . Passing to a further increasing choice if necessary, choose N_j such that

$$d(a_{N_j}^j, a_n^j) < \varepsilon_j \quad \text{for all } n \geq N_j.$$

Set $b_j = a_{N_j}^j$.

We claim that (b_j) is a Cauchy sequence in X . Fix $j \leq k$. Since $D(\eta_j, \eta_k) < \varepsilon_j$, we may choose N so large that

$$N \geq N_j, \quad N \geq N_k, \quad d(a_N^j, a_N^k) < 2\varepsilon_j.$$

The three-step relaxed triangle inequality gives

$$\begin{aligned} d(b_j, b_k) &= d(a_{N_j}^j, a_{N_k}^k) \\ &\leq s d(a_{N_j}^j, a_N^j) + s^2 d(a_N^j, a_N^k) + s^2 d(a_N^k, a_{N_k}^k) \\ &\leq (s + 2s^2)\varepsilon_j + s^2\varepsilon_k. \end{aligned}$$

This tends to 0 as $j, k \rightarrow \infty$, so $(b_j) \in \mathcal{C}$. Let

$$\beta = [(b_j)] \in \widehat{X}.$$

We first compare η_j with the particular representative $b = (b_n)$ of β . For fixed j and large $n \geq j$, compare a_n^j with $b_n = a_{N_n}^n$. Choose M large enough so that the tails of a^j and a^n beyond n and N_n are small and so that

$$d(a_M^j, a_M^n) < 2\varepsilon_j,$$

which is possible because $D(\eta_j, \eta_n) < \varepsilon_j$. Applying the three-step estimate,

$$d(a_n^j, b_n) \leq s d(a_n^j, a_M^j) + s^2 d(a_M^j, a_M^n) + s^2 d(a_M^n, a_{N_n}^n).$$

Taking the limsup as $n \rightarrow \infty$, the first and third terms vanish respectively by the Cauchy property of a^j and by the choice of N_n , while the middle term is bounded by $2s^2\varepsilon_j$. Hence

$$\limsup_{n \rightarrow \infty} d(a_n^j, b_n) \leq 2s^2\varepsilon_j.$$

Let $x^\beta = (x_n^\beta)$ be the globally chosen representative of the class β . Since $x^\beta \sim b$, the relaxed triangle inequality gives

$$D(\eta_j, \beta) = \limsup_{n \rightarrow \infty} d(a_n^j, x_n^\beta) \leq 2s^3\varepsilon_j.$$

Thus $\eta_j \rightarrow \beta$.

Finally, the original Cauchy sequence (ξ_k) also converges to β . If $k \geq k_j$, then

$$D(\xi_k, \beta) \leq s(D(\xi_k, \eta_j) + D(\eta_j, \beta)) \leq s(\varepsilon_j + 2s^3\varepsilon_j).$$

Letting $j \rightarrow \infty$ proves $\xi_k \rightarrow \beta$. Therefore $(\widehat{X}, D)$ is complete, and the theorem follows.

Verification and novelty notes

The proof solves exactly the existence problem in Remark 1.18. It does not assert uniqueness of the completion. The use of chosen representatives is essential to avoid the non-well-defined limit-distance obstruction in arbitrary b -metric spaces.

Local cheap-index search found no existing packet or attempt for arXiv:1802.02722. Local source search found the arbitrary-completion problem only in Remark 1.18 and found the known strong-b-metric completion theorem in the preceding paragraph. Web search on 2026-06-18 for exact phrases including “completion of an arbitrary b-metric space”, “Does every b-metric space admit a completion”, and “b-metric completion open problem” found the strong-b-metric answer arXiv:1503.08126 but no later full arbitrary-b-metric completion theorem.

Human-review recommendation

Review the dependence on the initial choice of representatives and the diagonal completeness argument. In particular, check the estimate proving (b_j) is Cauchy and the synchronized-limsup estimate proving $\eta_j \rightarrow \beta$. If these two steps pass review, this is a full solution to the source problem as stated.

References

- [1] S. Cobzas. *b-metric spaces, fixed points and Lipschitz functions*. arXiv:1802.02722, 2018.